Please check the examination details below before entering your candidate information

Candidate surname	Other names

Pearson Edexcel
Level 1/Level 2 GCSE (9–1)

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Half Past Paper – Week 13

Higher Tier

Total Marks

Q	Topic	Marks	Sparx Code
1	Position-to-term rules for arithmetic sequences	2	U498
3	Interpreting graphs of quadratic functions, cubic functions, reciprocal functions	2	U667,U980, U593
5	Finding the percentage an amount has been changed by	3	U278
7	Interpreting stem-and-leaf diagrams	3	U909
9	Using standard form with positive indices and negative indices	2	U330,U534
11a	Using the correct order of operations, Calculating with roots and powers	2	U976,U851
11b	Indices of the form 1/a	2	U985
11c	Simplifying expressions using index laws	2	U662
13	Calculating with density	3	U910
15	Equations of parallel and perpendicular lines	3	U898
17	Changing the subjects of formulae, Expanding and Factorising single brackets	4	U556,U179, U365
19a	Substituting into functions	1	U637
19b	Substituting into composite functions	2	U895
19c	Finding inverse functions	2	U996
21	Solving geometric problems using vectors, Identifying parallel vectors	4	U781,U660
23	Changing ratios	3	U865
	Total	40	

Answer ALL questions.

Write your answers in the spaces provided.

You must write down all the stages in your working.

1 The first five terms of an arithmetic sequence are

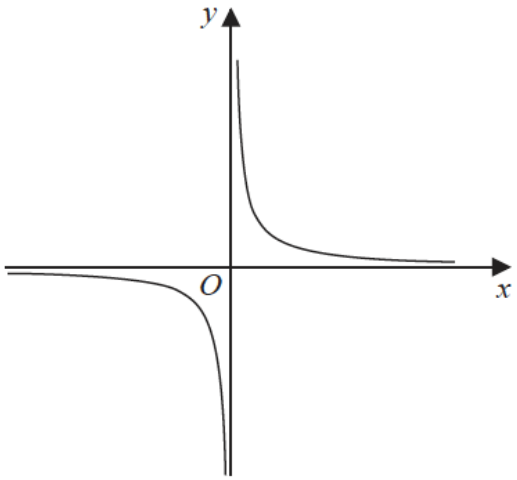
1 4 7 10 13

Write down an expression, in terms of n, for the nth term of this sequence.

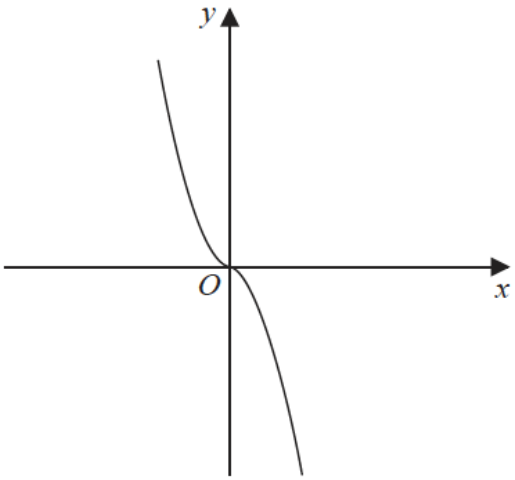
.....

(Total for Question 1 is 2 marks)

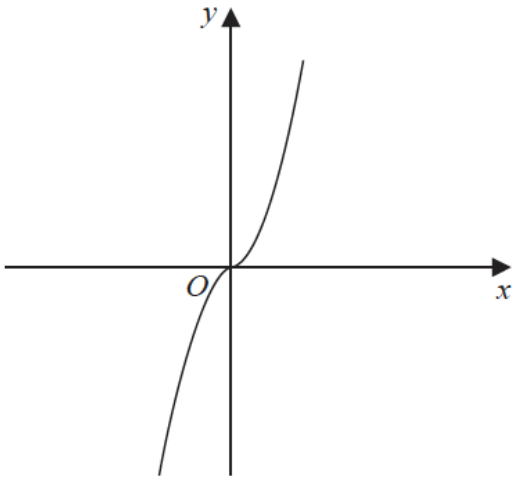
3 The diagram shows four graphs.



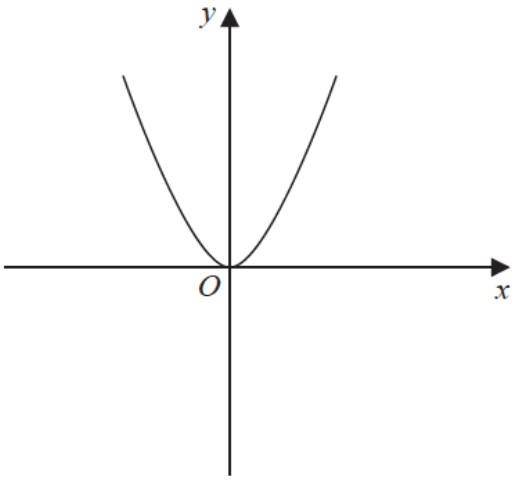
Graph **A**



Graph **B**



Graph **C**



Graph **D**

Each of the equations in the table is the equation of one of the graphs.

Complete the table.

Equation	**Letter of graph**
$y = -x^3$	
$y = x^3$	
$y = x^2$	
$y = \frac{1}{x}$	

(Total for Question 3 is 2 marks)

5 Sean pays £10 for 24 chocolate bars.

He sells all 24 chocolate bars for 50p each.

Work out Sean's percentage profit.

..... %

(Total for Question 5 is 3 marks)

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7 The table shows information about the heights, in cm, of a group of Year 9 girls.

least height	150 cm
median	165 cm
greatest height	170 cm

This stem and leaf diagram shows information about the heights, in cm, of a group of 15 Year 9 boys.

15	8 9 9
16	4 5 7 7 8
17	0 3 4 4 7
18	0 2

Key: 15 | 8 represents 158 cm

Compare the distribution of the heights of the girls with the distribution of the heights of the boys.

.....

.....

.....

.....

(Total for Question 7 is 3 marks)

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9 Write these numbers in order of size.
Start with the smallest number.

6.72×10^5 67.2×10^{-4} 672×10^4 0.000 672

....

(Total for Question 9 is 2 marks)

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11 (*a*) Find the value of $\sqrt[4]{81 \times 10^8}$

.....

(2)

(*b*) Find the value of $64^{-\frac{1}{2}}$

.....

(2)

(*c*) Write $\frac{3^n}{9^{n-1}}$ as a power of 3

.....

(2)

(Total for Question 11 is 6 marks)

13 Liquid **A** and liquid **B** are mixed to make liquid **C**.

Liquid **A** has a density of 70 kg/m^3
Liquid **A** has a mass of 1400 kg

Liquid **B** has a density of 280 kg/m^3
Liquid **B** has a volume of 30 m^3

Work out the density of liquid **C**.

..... kg/m^3

(Total for Question 13 is 3 marks)

15 The straight line L_1 has equation $y = 3x - 4$
The straight line L_2 is perpendicular to L_1 and passes through the point (9, 5)

Find an equation of line L_2

.....

(Total for Question 15 is 3 marks)

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17 Make f the subject of the formula $d = \dfrac{3(1 - f)}{f - 4}$

.....

(Total for Question 17 is 4 marks)

19 f and g are functions such that

$$f(x) = \frac{12}{\sqrt{x}} \qquad \text{and} \qquad g(x) = 3(2x + 1)$$

(*a*) Find g(5)

.....

(1)

(*b*) Find gf(9)

.....

(2)

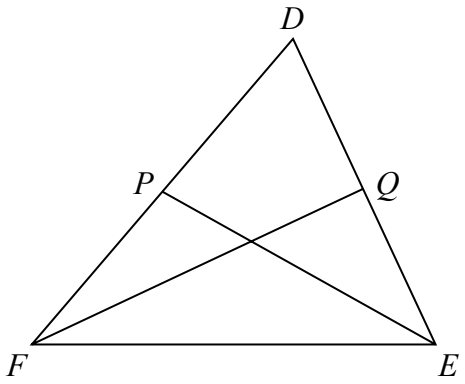
(*c*) Find $g^{-1}(6)$

.....

(2)

(Total for Question 19 is 5 marks)

21 *DEF* is a triangle.



P is the midpoint of *FD*.
Q is the midpoint of *DE*.

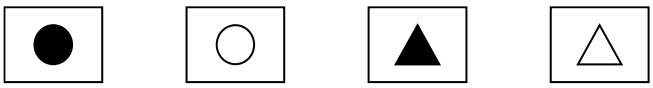
$\overrightarrow{FD}$ = **a** and $\overrightarrow{FE}$ = **b**

Use a vector method to prove that *PQ* is parallel to *FE*.

(Total for Question 21 is 4 marks)

23 There are four types of cards in a game.

Each card has a black circle or a white circle or a black triangle or a white triangle.



number of cards with a black shape	:	number of cards with a white shape	=	3 : 5
number of cards with a circle	:	number of cards with a triangle	=	2 : 7

Express the total number of cards with a black shape as a fraction of the total number of cards with a triangle.

.....

(Total for Question 23 is 3 marks)

TOTAL FOR PAPER IS 40 MARKS